USN | | | | | | | | | | | [23MSCH1H01]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY - 2025

INORGANIC CHEMISTRY I

Time: 03 HOURS **Maximum Marks:50**

		Blooms Level	CO	Marks
	SECTION – A			
1.	**Answer any FOUR of the following questions.**			**4×5=20**
a.	Give an account of cyclic silicates with an example.	2	2	**05**
b.	Sketch and explain the molecular orbital energy level diagram for F_2 molecule.	6	3	**05**
c.	What is the difference between a perfect metal and a superconductor? Draw resistivity vs. temperature plot of a superconductor.	3	4	**05**
d.	Differentiate Lewis and Lux-Flood acid-base concepts.	3	5	**05**
e.	What is radioactivity? Explain alpha, beta and gamma decay processes.	2	6	**05**
	SECTION – B			
2.	**Answer any TWO of the following questions.**			**2×9=18**
a.	**i.** Explain the anomalous behavior of the first element in each group of the p-block (e.g., boron, carbon, nitrogen, oxygen). **ii.** How does this deviation arise and affects their chemical properties compared to the other members of the group? Provide examples to illustrate your answer.	5	1	**09**
b.	Draw a molecular orbital energy-level diagram for N_2 and F_2. Why is the bond strength of the N_2 molecule greater than that of the F_2 molecule?	3	3	**09**
c.	What is a nuclear chain reaction and how can it be controlled? Discuss the nuclear reactions which are involved in a nuclear reactor.	2	6	**09**
	SECTION – C			
3.	**Answer the following question.**			**1x12=12**
a.	**i.** Derive the expression for electrical conductivity of a metal using free electron theory.	6	4	**04**
	ii. Explain the Bronsted-Lowry acid-base concept and Arrhenius acid-base concepts with examples.	4	5	**04**
	iii. What are Sulphur-Nitrogen compounds? Give two examples.	1	6	**04**